

Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the **Credit Card Approval Prediction** project codebase. The project follows a modular architecture with separate components for data preprocessing, machine learning model training, prediction, frontend, and backend. Meaningful naming conventions, proper documentation, and reusable modules improve maintainability and future enhancements.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Code follows consistent indentation and formatting.
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for frontend, backend, dataset, and ML model.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	variables like income, creditScore, prediction Result are descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as trainModel(), predictApproval(), validate Input() are meaningful.
5	Code Comments	Inline and block comments explain logic	Yes	Important sections include comments for better understanding.
6	Modular Design	Code is split into reusable functions	Yes	ML model, preprocessing, API, and UI are developed as separate modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Common logic is reused, avoiding unnecessary duplication.
8	Error Handling	Exceptions and errors are handled	Yes	Input validation and exception handling improve reliability.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python, Pandas	Model Training & Prediction	High
2	Machine Learning Prediction Model	Python, Scikit-learn	Credit Card Approval Prediction	High
3	REST API Module	Flask / FastAPI	Frontend and Backend Communication	High
4	User Authentication Module	HTML, CSS, JavaScript	Login and Admin/User Access	Medium
5	Prediction Result Dashboard	React.js	Displays Prediction Results and Reports	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized folder structure with clear separation of modules.
Readability	5	Meaningful naming conventions, formatting, and comments improve readability.
Reusability	4	Core modules are reusable across different parts of the application.
Documentation / Comments	4/5	Adequate comments and documentation are provided for key functionalities.
Overall Score	4	The project follows good coding standards, modular design, and reusable architecture, making it maintainable and scalable.